

**Course Code: BET-C620**  
**Course Name: Embedded Systems and IoT Applications**

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| <b>MM: 100</b><br><b>Time: 3 Hr.</b><br><b>L T P</b><br><b>3 1 0</b> | <b>Sessional:30</b><br><b>ESE:70</b><br><b>Credit :4</b> |
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| <b>Prerequisites:</b>     | Microprocessor & interfacing                                                                                                                                                                                                                                                                                                                                                 |
| <b>Objectives:</b>        | <ol style="list-style-type: none"> <li>1. Understand the Fundamentals of Embedded Systems.</li> <li>2. Develop Programming Skills for Embedded Systems.</li> <li>3. To explore serial, parallel, and wireless communication protocols.</li> <li>4. To develop skills for real-world peripheral interfacing.</li> <li>5. Understanding of IoT and its applications</li> </ol> |
| <b>Course Coordinator</b> | Mr. ANUJ KUMAR SHARMA                                                                                                                                                                                                                                                                                                                                                        |

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| <b>NOTE:</b> | The question paper shall consist of two sections (Section-A and Section-B). Section-A shall contain of ten (10) short answer type questions of six (06) mark each and student shall be required to attempt any five (05) questions. Section-B shall contain eight (08) long answer type questions of ten (10) marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus. |
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| UNIT   | Module   | Course Content                                                                                                                                                                                                                                                                                                                                            | No. of Hours | POs mapped           | PSOs mapped      |
|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|------------------|
| UNIT-1 | Module-1 | Embedded systems and its applications, Embedded Operating system, Design parameters of an embedded system and its significance, design life cycle, tools introduction, hardware and software partitioning and co-design. Open collector outputs Tristate outputs I/O sinking and Sourcing, PLD's, Watchdog Timers, Hardware design and development.       | 06           | PO1/<br>PO2/<br>PO3  | PSO1/<br>PSO2/.. |
| UNIT-2 | Module-2 | Microcontroller families and classifications (8-bit, 16-bit, 32-bit), 8051 Microcontroller, AVR Microcontroller, Features of ATmega328 used in Arduino, PIC Microcontrollers, ARM Cortex-M Series (e.g., STM32), ESP32 / ESP8266 (Wi-Fi-enabled microcontrollers), Programming via Arduino IDE and ESP-IDF.                                               | 08           | PO1/<br>PO2/<br>PO3  | PSO1/<br>PSO2/.. |
| UNIT-3 | Module-3 | Introduction to Microcontrollers and Microprocessors, Embedded versus external memory devices, CISC and RISC processors, Harvard and Von Neumann Architectures. 8051 Microcontrollers-Assembly language, architecture, registers, Addressing modes, Instruction set, I/O ports and memory organization Interrupts Timer/counter and serial communication. | 12           | PO2/<br>PO3/<br>PO4/ | PSO1/<br>PSO2/.. |
| UNIT-4 | Module-4 | Communication basics, Microprocessor Interfacing I/O Addressing, Direct memory access, Arbitration, multilevel bus architecture, Serial protocols, Parallel protocols and wireless                                                                                                                                                                        | 10           | PO2/<br>PO4/<br>PO5  | PSO1/<br>PSO2/.. |

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|                    |          | protocols. Real world Interfacing: LCD, Stepping Motor, ADC, DAC, LED, Push Buttons, Key board, Latch Interconnection, PPI.                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |                     |                  |
| UNIT-5             | Module-5 | Introduction to Internet of Things (IoT), components of an IoT system including sensors, actuators, microcontrollers, and cloud platforms; IoT system architecture, overview of IoT communication models such as device-to-device, device-to-cloud, and gateway-based communication; IoT communication technologies including Wi-Fi for local high-speed connections, Bluetooth Low Energy (BLE) for short-range low-power communication, Zigbee for mesh networking, LoRaWAN for long-range low-power applications, and NB-IoT for cellular-based IoT solutions. | 04 | PO2/<br>PO4/<br>PO5 | PSO1/<br>PSO2/.. |
| Total No. of Hours |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 40 |                     |                  |

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| <b>Learning Outcomes:</b> | <p>After completing this course, the student will be able to</p> <ol style="list-style-type: none"> <li>1. <b>Explain</b> the fundamentals of embedded systems and their applications in real-world scenarios.</li> <li>2. <b>Develop</b> basic embedded programs using 8051, AVR (ATmega328), PIC, ARM Cortex-M, and ESP32/ESP8266 microcontrollers.</li> <li>3. <b>Describe</b> the structure and functions of a Real-Time Operating System (RTOS), including tasks, semaphores, shared data, queues, and mailboxes.</li> <li>4. <b>Compare</b> serial, parallel, and wireless communication protocols for use in embedded applications.</li> <li>5. <b>Interface</b> real-world devices (LCD, stepper motors, ADC/DAC, LEDs, push buttons, keyboards) with microcontrollers.</li> </ol> |
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**Suggested books:**

| S. No. | Name of Authors /Books /Publisher                                                                                                            | Year of Publication |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1.     | Muhammad Ali Mazidi and Janice Gillispie " <i>The 8051 Microcontroller and embedded systems</i> " ISBN: 978-0131194021                       | 1999                |
| 2.     | Tony Givargis Frank Vahid " <i>Embedded System Design: A Unified Hardware / Software Introduction</i> ", IV, McGraw-Hill, ISBN-9780071371766 | 2006                |
| 3.     | Kenneth Hintz, Daniel Tabak " <i>Microcontrollers (Architecture, Implementation &amp; Programming)</i> " Tata McGraw-Hill,                   | 2005                |
| 4.     | Sampath Kr " <i>Microcontrollers &amp; Embedded Systems 2nd Edition</i> " KatsonBooks                                                        | 206                 |

|      | Learning Outcome 1 | Learning Outcome 2 | Learning Outcome 3 | Learning Outcome 4 | Learning Outcome 5 |
|------|--------------------|--------------------|--------------------|--------------------|--------------------|
| PO1  | Y                  | N                  | Y                  | Y                  | Y                  |
| PO2  | Y                  | Y                  | N                  | Y                  | Y                  |
| PO3  | Y                  | N                  | Y                  | Y                  | Y                  |
| PO4  | N                  | Y                  | N                  | N                  | Y                  |
| PO5  | Y                  | N                  | Y                  | Y                  | Y                  |
| PO6  | Y                  | Y                  | N                  | Y                  | Y                  |
| PO7  | N                  | N                  | N                  | N                  | Y                  |
| PO8  | Y                  | Y                  | N                  | Y                  | Y                  |
| PO9  | Y                  | N                  | Y                  | N                  | Y                  |
| PO10 | Y                  | Y                  | N                  | Y                  | Y                  |
| PO11 | Y                  | N                  | N                  | Y                  | Y                  |
| PO12 | Y                  | Y                  | N                  | Y                  | Y                  |

|      | Learning Outcome 1 | Learning Outcome 2 | Learning Outcome 3 | Learning Outcome 4 | Learning Outcome 5 |
|------|--------------------|--------------------|--------------------|--------------------|--------------------|
| PSO1 | N                  | Y                  | N                  | N                  | N                  |
| PSO2 | Y                  | N                  | Y                  | Y                  | N                  |
| PSO3 | Y                  | N                  | Y                  | Y                  | N                  |
| PSO4 | N                  | N                  | Y                  | Y                  | Y                  |